

Sampler-independent integrability horizons and bounded unbiased estimators for branching representations of semilinear PDEs

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Abstract

Branching (coding-tree) Monte Carlo methods represent the solution of a semilinear parabolic PDE as the expectation of a random functional of a marked branching diffusion. Their use is limited by integrability: beyond a short horizon the functional ceases to have a finite variance, and the literature provides sufficient conditions only. We show that for a fixed tree expansion the first absolute moment of the estimator does not depend on the sampling law (lifetime distribution, branching probabilities, adaptive choices) and equals the minimal solution of a positive *majorant system*. Conversely, whenever this majorant is finite we construct, from any supersolution of the majorant system, an unbiased estimator whose output is bounded by the supersolution, and, when the supersolution has a multiplicative slack $\vartheta > 1$, whose expected number of generated nodes is at most $\vartheta/(\vartheta - 1)$ uniformly in the horizon. In particular, the supremum over all samplers of the horizon of finite variance equals the absolute-integrability horizon, and it is attained with bounded output. We compute this horizon exactly for constant data, including its endpoint behaviour, classify all-horizon integrability on compact tori, exhibit a Fujita-type dimension threshold on $\mathbb{R}^d$, and show that standard fixed-rate samplers lose finite variance strictly earlier: for flat Allen–Cahn data at most 0.9955 against the intrinsic horizon $(3\pi - 2\log 3)/5 \approx 1.4455$, and with the uniform tuple choice of standard implementations only $0.9955/\sqrt{d+1}$ in dimension d . The node-weighted mass is sampler independent as well, which makes the work of bounded-output sampling near the horizon exactly $\Theta((\tau - T)^{-1})$ for constant data. Numerical experiments on Allen–Cahn data in dimensions 1 and 100 confirm certified confidence intervals up to 94% of the guaranteed horizon, where fixed-rate samplers show heavy-tailed failure.

1 Introduction

Probabilistic representations of semilinear PDEs by branching diffusions go back to Skorokhod, Watanabe and McKean [15]; their use as Monte Carlo schemes free of spatial grids was developed in [1, 8–10] and extended to nonlinearities involving derivatives of arbitrary order through *coding trees* [18, 19]. A particle carries a *code* (an operator applied to the unknown solution), lives for a random time, and branches into a tuple of codes chosen from a mechanism derived from the chain rule. The estimator H is a product of importance weights and terminal values along the tree, and $u = \mathbb{E}H$.

The central practical difficulty is integrability. The random functional H is a product over a random tree of weights that are typically larger than one; for all but short horizons its variance is infinite, and the Monte Carlo average then converges slowly and erratically, with heavy-tail artefacts that masquerade as bias. Existing analyses provide sufficient conditions: comparison with an ordinary differential equation [8, Thm. 3.2], growth conditions on jets of the data [11, 12], or small-maturity assumptions [10, 18]. These conditions depend on the lifetime law and branching

probabilities, which are free parameters of the method, and it has remained unclear which part of the “short time” restriction is intrinsic to the representation and which part is an artefact of the sampler.

Contributions. We fix the tree *expansion* (the code set, mechanism, coefficients and terminal functions) and vary everything else.

1. **Sampler independence (Theorem 3.1).** For every sampling law whose support covers the nonzero trees, $\mathbb{E}|H| = W_{\text{Id}}(T, x)$, the minimal solution of a positive majorant system obtained by replacing every coefficient and terminal value by its absolute value. No choice of clock or branching probabilities can extend the horizon on which $\mathbb{E}|H| < \infty$.
2. **Attainability with bounded output and horizon-free work (Theorem 4.2).** Tilting the tree law by any supersolution $\widehat{W}$ of the majorant system yields an unbiased estimator with $|H| \leq \widehat{W}_{\text{Id}}(T, x)$ almost surely; a multiplicative slack $\vartheta > 1$ in the supersolution inequality turns into a geometric bound on the number of generated nodes, $\mathbb{E}N \leq \vartheta/(\vartheta - 1)$, independent of T . Consequently the supremum over samplers of the finite-variance horizon equals the absolute-integrability horizon (Corollary 4.3). Sup-norm majorants, which previously served as sufficient conditions [8], become constructive: they define the sampler.
3. **Exact horizons.** For constant data the horizon is $\tau = \int_0^R dz/\Phi(z)$ with Φ the absolute Taylor series of the nonlinearity; we identify the endpoint behaviour and the exact effect of the slack, $W_{\text{Id}}^\vartheta(t) = \vartheta|r| + Z(\vartheta^2 t)$ (Theorem 5.1). On compact tori, all-horizon integrability holds iff the nonlinearity is affine on the range of the data (analytic case, Theorem 5.7); on $\mathbb{R}^d$ with small Gaussian data and quadratic absorption it holds iff $d \geq 5$ (Theorem 5.8).
4. **The gap for standard samplers (Proposition 5.4).** With a fixed exponential clock, the second moment is governed by the squared-jet series Φ_2 and finite exactly for $T < \lambda^{-1} \log(1 + \lambda^2 p \tau_2)$; optimizing λ gives at most $0.80474\sqrt{\tau_2}$, which for flat Allen–Cahn data is $0.9955 < 1.4455 = \tau$. For two-term nonlinearities of degree p the attainable fraction of τ is an explicit constant tending to 0 as $p \rightarrow \infty$ (Remark 5.6). In dimension d , uniform tuple choice multiplies the limit by $(d + 1)^{-1/2}$ (Corollary 5.5), whereas the sup-majorant tilt is dimension free (Proposition 4.6).
5. **Work.** The node-weighted mass $\mathbb{E}[N|H|]$ is also sampler independent, which gives a lower bound on the work of any sampler with bounded output. For constant data, sampling with output $O(W)$ costs exactly $\Theta((\tau - T)^{-1})$ nodes near the horizon (Corollary 5.2).

Relation to prior work. Zero-variance importance sampling [13], Doob transforms of branching processes and Boltzmann samplers for the trees of $Y' = \Phi(Y)$ [2, 4] are classical, and the cancellation behind Theorem 3.1 is implicit in [9]. For integral equations with power nonlinearity $\varphi = K\varphi^n + h$, branching estimators go back to Ermakov [5, 6], and variance criteria through majorant equations and “value modelling” (densities multiplied by the solution) were developed by Mikhailov and coauthors [16, 17]; see also [20] for the parabolic Cauchy problem. Medvedev and Mikhailov [16, §6] note that an ideal zero-variance version of the nonlinear estimator had not been studied; for the *collision* estimator, where every node contributes h and keeps branching, the ideal transition density $k\varphi^n/(\varphi - h)$ generates infinite trees, and their Theorem 4 yields zero variance only under $1 - h/\varphi < 1/n$. The tilt of Section 4 is value modelling in the *absorption* form, where each node chooses between a leaf and a branch in proportion to their masses; there the ideal tilt always produces almost surely finite trees with $|H| = W$ (Theorem 4.2(f)), hence zero variance for nonnegative problems,

without any condition. The underlying principle is theirs; we do not claim it. The majorant ODE of [8, Thm. 3.2] is the spatially constant instance of our system, used there as an analysis device for a sufficient L^1 condition, with the branching law chosen independently. Branching-particle proofs of Fujita-type blow-up are classical [3, 14]; Theorem 5.8 applies such methods, together with [7], to the moment system of a coding tree. What we believe is new is the formulation for derivative-coded PDE trees in which the sampler-independent horizon is an *equivalence* (Corollary 4.3), the horizon-free work bound obtained from a multiplicative slack (Theorem 4.2(c)), and the exact horizons and obstructions of Theorems 5.1–5.8 and Proposition 5.4. **[TODO: check [17] (book) for an absorption-form ideal estimator or a slack/work bound; [16] checked in full text: collision form only, condition as above, work discussed only through $n\|K\| < 1$; extend the search to neutron-transport weight windows].**

2 Tree expansions and the majorant system

Let $\mathbf{X}$ be $\mathbb{R}^d$ or the flat torus $\mathbb{T}^d = \mathbb{R}^d/\mathbb{Z}^d$, let $p_t(y, z)$ be the heat kernel of $L = \frac{1}{2}\Delta$ and P_t its semigroup. Time t denotes *remaining* time: we consider

$$\partial_t u = \frac{1}{2}\Delta u + f(u), \quad u(0, \cdot) = v, \quad (1)$$

which is the terminal-value problem $\partial_s u + \frac{1}{2}\Delta u + f(u) = 0$, $u(T) = v$, after $t = T - s$.

Definition 2.1 (Expansion). An *expansion* consists of a countable code set $\mathcal{C} \ni \text{Id}$, measurable terminal functions $g_c : \mathbf{X} \rightarrow \mathbb{R}$, and for each $c \in \mathcal{C}$ a finite list $\mathcal{M}(c) = \{(\kappa_j, (c_{j,1}, \dots, c_{j,m_j}))\}_j$ of nonzero real coefficients and code tuples.

Example 2.2 (Semilinear coding tree [18]). For (1) with f smooth, take codes Id , F_k ($k \geq 0$, standing for $f^{(k)}(u)$) and D_i ($i \leq d$, standing for $\partial_i u$), terminal functions $g_{\text{Id}} = v$, $g_{F_k} = f^{(k)}(v)$, $g_{D_i} = \partial_i v$, and

$$\text{Id} \rightarrow (F_0), \quad F_k \rightarrow (F_0, F_{k+1}), \quad F_k \xrightarrow{-1/2} (D_i, D_i, F_{k+2}), \quad D_i \rightarrow (F_1, D_i),$$

with coefficient 1 unless indicated. The tuples encode $(\partial_t - L)f^{(k)}(u) = f^{(k+1)}(u)f(u) - \frac{1}{2}f^{(k+2)}(u)|\nabla u|^2$ and $(\partial_t - L)\partial_i u = f'(u)\partial_i u$.

Trees and the canonical measure. A completed tree rooted at (c, t, y) is either a *leaf* with terminal position $z \in \mathbf{X}$, or a *branch* consisting of a lifetime $\sigma \in (0, t)$, a branch position z , an index j , and one completed tree rooted at $(c_{j,i}, t - \sigma, z)$ for each $i \leq m_j$; all children share the position z . The canonical measure $\mu_{c,t,y}$ on finite completed trees is defined recursively by

$$\mu_{c,t,y} = \delta_{\text{leaf}} \otimes p_t(y, z) \, dz + \sum_j \mathbf{1}_{(0,t)}(\sigma) \, d\sigma \otimes p_\sigma(y, z) \, dz \otimes \bigotimes_i \mu_{c_{j,i}, t-\sigma, z},$$

on trees of depth at most n for each n (induction on n). The signed weight $a(\theta)$ of a tree is the product of $g_c(z)$ over leaves and κ_j over branches; $N(\theta)$ denotes its number of nodes.

Definition 2.3 (Majorant). For $\vartheta \geq 1$ set $W_c^\vartheta(t, y) = \int \vartheta^N |a| \, d\mu_{c,t,y} \in [0, \infty]$ and $W = W^1$. When $W_c(t, y) < \infty$ put $U_c(t, y) = \int a \, d\mu_{c,t,y}$.

Lemma 2.4. W^ϑ is the minimal $[0, \infty]$ -valued solution of

$$W_c^\vartheta(t, y) = \vartheta \left[P_t |g_c|(y) + \sum_j |\kappa_j| \int_0^t P_\sigma \left[\prod_i W_{c_{j,i}}^\vartheta(t - \sigma, \cdot) \right] (y) d\sigma \right]. \quad (2)$$

If $W_{\text{Id}}(t, y) < \infty$ then U satisfies the signed version of (2) (with $\vartheta = 1$, g_c , κ_j) at (Id, t, y) , and at almost every state reached from it.

Proof. Tonelli's theorem applied to the recursive definition gives (2). Let $W^{(n)}$ be the integral over trees of depth at most n ; then $W^{(n)} \uparrow W^\vartheta$, and if $V \geq 0$ solves (2) then $W^{(n)} \leq V$ by induction on n . The signed statement follows by Fubini under absolute integrability. $\square$

Proposition 2.5 (Correspondence). *In Example 2.2 let f be a polynomial and $v \in C_b^2(\mathbf{X})$. If W_c is bounded on $[0, T] \times \mathbf{X}$ for every code c , then (1) has a unique bounded classical solution on $[0, T]$, and $U_{\text{Id}} = u$ there.*

The proof (Appendix A) is the finite-system uniqueness argument of [8, Thm. 3.2]; compare [18, Thm. 4.2] and [11], where integrability is assumed or derived from sufficient conditions. In Theorem 4.2 it suffices that the supersolution is bounded, since $W \leq \widehat{W}$.

3 Samplers and sampler independence

A *sampler with cemetery* for (T, x) is a probability law Q on finite completed trees rooted at (Id, T, x) together with a cemetery point $\dagger$. On a tree, the estimator is $H = a \, d\mu / dQ$; on $\dagger$, $H = 0$. We call Q *supported* if $dQ / d\mu > 0$ μ -a.e. on $\{a \neq 0\}$. The fixed-rate samplers of [10, 18] (exponential lifetimes of rate λ , tuple probabilities $q_j > 0$, Brownian positions) are supported samplers without cemetery, with H the familiar product of $\kappa_j / (q_j \lambda e^{-\lambda\sigma})$ over branches and $g / e^{-\lambda t}$ over leaves. Adaptive choices (depending on the history of the tree) are allowed; only the joint density on complete trees enters.

Theorem 3.1 (Sampler independence). *Let $W_N(T, x) = \int N|a| \, d\mu_{\text{Id}, T, x}$. For every supported sampler,*

$$\mathbb{E}_Q |H| = W_{\text{Id}}(T, x), \quad \mathbb{E}_Q [N | H] = W_N(T, x).$$

Consequently $\mathbb{E}_Q H^2 \geq W_{\text{Id}}(T, x)^2$; if $W_{\text{Id}}(T, x) = \infty$ no supported sampler of this expansion has a finite first absolute moment; and every supported sampler with $|H| \leq M$ a.s. satisfies $\mathbb{E}_Q N \geq W_N(T, x) / M$. Moreover $W_N = \lim_{\vartheta \downarrow 1} (W_{\text{Id}}^\vartheta - W_{\text{Id}}) / (\vartheta - 1)$, and $W_N \leq W_{\text{Id}}^\vartheta / (\vartheta - 1)$ for every $\vartheta > 1$.

Proof. For any measurable $\psi \geq 0$ on trees, $\mathbb{E}_Q [\psi | H] = \int_{\{a \neq 0\}} \psi |a| \frac{d\mu}{dQ} dQ = \int \psi |a| d\mu$; take $\psi = 1$ and $\psi = N$. Jensen's inequality gives the second-moment bound, and $\mathbb{E}_Q [N | H] \leq M \mathbb{E}_Q N$ the work bound. The map $\vartheta \mapsto W_{\text{Id}}^\vartheta = \int \vartheta^N |a| d\mu$ is convex with right derivative W_N at 1 (monotone convergence of the difference quotients $(\vartheta^N - 1) / (\vartheta - 1) \downarrow N$), so $W_{\text{Id}}^\vartheta \geq W_{\text{Id}} + (\vartheta - 1) W_N$. $\square$

4 Tilted samplers

Definition 4.1. Let $\vartheta \geq 1$ and let $h_c \geq |g_c|$ be measurable envelopes. A measurable $\widehat{W} : \mathcal{C} \times (0, T] \times \mathbf{X} \rightarrow [0, \infty]$ is a ϑ -*supersolution* if for all (c, t, y)

$$\widehat{W}_c(t, y) \geq \vartheta \left[L_c(t, y) + \sum_j |\kappa_j| B_{c,j}(t, y) \right], \quad L_c = P_t h_c, \quad B_{c,j}(t, y) = \int_0^t \int p_\sigma(y, z) \prod_i \widehat{W}_{c_{j,i}}(t - \sigma, z) dz d\sigma, \quad (3)$$

with the convention $0 \cdot \infty = 0$.

The tilted sampler. Define a recursive procedure $S(c, t, y)$ with values in $[-1, 1]$:

1. with probability $L_c/\widehat{W}_c$: leaf; draw z with density $p_t(y, z)h_c(z)/L_c(t, y)$ and return $g_c(z)/h_c(z)$;
2. with probability $|\kappa_j|B_{c,j}/\widehat{W}_c$: branch j ; draw (σ, z) with density $p_\sigma(y, z) \prod_i \widehat{W}_{c_{j,i}}(t-\sigma, z)/B_{c,j}(t, y)$ and return $\text{sgn}(\kappa_j) \prod_i S(c_{j,i}, t-\sigma, z)$, evaluating the children one after another;
3. otherwise: kill, i.e. stop the whole computation and output 0.

The estimator is $H = \widehat{W}_{\text{Id}}(T, x) S(\text{Id}, T, x)$ (and $H = 0$ if $\widehat{W}_{\text{Id}}(T, x) = 0$).

Theorem 4.2. *Let $\widehat{W}$ be a ϑ -supersolution with $\widehat{W}_{\text{Id}}(T, x) < \infty$.*

- (a) *Almost surely every visited state has $0 < \widehat{W} < \infty$.*
- (b) *$|H| \leq \widehat{W}_{\text{Id}}(T, x)$ almost surely.*
- (c) *If $\vartheta > 1$, the number N_{gen} of generated nodes satisfies $\mathbb{P}(N_{\text{gen}} > n) \leq \vartheta^{-n}$; in particular $\mathbb{E}N_{\text{gen}} \leq \vartheta/(\vartheta - 1)$.*
- (d) *If the procedure stops almost surely (always when $\vartheta > 1$), then $W_{\text{Id}}(T, x) \leq \widehat{W}_{\text{Id}}(T, x)$, the sampler is supported, and $\mathbb{E}H = U_{\text{Id}}(T, x)$.*
- (e) *$\mathbb{E}H^2 \leq \widehat{W}_{\text{Id}}(T, x) W_{\text{Id}}^{(h)}(T, x)$, where $W^{(h)}$ is the majorant with h in place of $|g|$; with $h = |g|$ equality holds, and $\mathbb{E}H^2 \leq (\widehat{W}_{\text{Id}}/W_{\text{Id}}) \inf_Q \mathbb{E}_Q H^2$ over supported samplers Q .*
- (f) *If $W_{\text{Id}}(T, x) < \infty$, the choice $\widehat{W} = W$, $h = |g|$, $\vartheta = 1$ never kills, stops almost surely, and gives $|H| = W_{\text{Id}}(T, x)$ a.s.*

Proof. (a) At a state with $\widehat{W}_c \in (0, \infty)$ the branch density has total mass $B_{c,j} \leq \widehat{W}_c/\vartheta < \infty$, so the set of (σ, z) at which some child has $\widehat{W} = \infty$ while the product is positive is null, and where the product vanishes the density vanishes. The root is covered by assumption. (b) By induction on the finite tree, $S \in [-1, 1]$. (c) Nodes are generated one at a time. Conditionally on everything generated before, the current node is killed with probability $1 - (L_c + \sum_j |\kappa_j|B_{c,j})/\widehat{W}_c \geq 1 - \vartheta^{-1}$ by (3). Generating an $(n+1)$ st node requires n non-kills. (d) The Q -density of a completed, non-killed tree θ is a product of step probabilities and conditional densities: a leaf contributes $p_t h_c/\widehat{W}_c$, a branch contributes $|\kappa_j|p_\sigma \prod_i \widehat{W}_{c_{j,i}}/\widehat{W}_c$. Every non-root $\widehat{W}$ appears once in a numerator and once in a denominator, hence

$$\frac{dQ}{d\mu}(\theta) = \frac{|a_h(\theta)|}{\widehat{W}_{\text{Id}}(T, x)}, \quad a_h = \prod_{\text{branches}} \kappa_j \prod_{\text{leaves}} h_c(z),$$

which is positive where $a \neq 0$. On θ , $H = \widehat{W}_{\text{Id}} \prod \text{sgn}(\kappa_j) \prod g_c/h_c = a \, d\mu/dQ$. Hence $\int |a| \, d\mu \leq \int |a_h| \, d\mu = \widehat{W}_{\text{Id}} Q(\text{completed}) \leq \widehat{W}_{\text{Id}}$, and, the procedure stopping a.s., $\mathbb{E}H = \int a \, d\mu = U_{\text{Id}}(T, x)$. (e) $\mathbb{E}H^2 \leq \widehat{W}_{\text{Id}}^2 Q(\text{completed}) = \widehat{W}_{\text{Id}} \int |a_h| \, d\mu$, with equality if $|g/h| = 1$; combine with Theorem 3.1. (f) With $\widehat{W} = W$, (3) holds with equality by Lemma 2.4, so no kill occurs, and completed trees have Q -mass $\int |a| \, d\mu/W_{\text{Id}} = 1$. $\square$

Corollary 4.3 (Sampler-independent horizon). *For fixed (T, x) the following are equivalent: (i) some supported sampler has $\mathbb{E}|H| < \infty$; (ii) every supported sampler has $\mathbb{E}|H| < \infty$; (iii) $W_{\text{Id}}(T, x) < \infty$; (iv) some supported sampler that stops almost surely has $|H| = W_{\text{Id}}(T, x)$ a.s. In particular the supremum over supported samplers of the horizons of finite variance equals the horizon of absolute integrability, and it is attained with bounded output.*

Corollary 4.4 (Horizon-free work). *If $W_{\text{Id}}^\vartheta(T, x) < \infty$ for some $\vartheta > 1$, tilting by W^ϑ with $h = |g|$ gives an unbiased estimator with $|H| \leq W_{\text{Id}}^\vartheta(T, x)$, $\mathbb{E}H^2 = W_{\text{Id}}^\vartheta(T, x)W_{\text{Id}}(T, x)$ and $\mathbb{E}N_{\text{gen}} \leq \vartheta/(\vartheta - 1)$.*

Proof. W^ϑ satisfies (3) with equality by Lemma 2.4; apply Theorem 4.2. $\square$

Remark 4.5 (Work-output trade-off). By Theorem 3.1, every supported sampler with output bound M needs $\mathbb{E}N \geq W_N/M$, so the product of the output bound and the expected work is at least W_N , whatever the sampler. The tilt of Corollary 4.4 achieves $M \mathbb{E}N \leq W_{\text{Id}}^\vartheta \vartheta/(\vartheta - 1)$. Section 5 shows that for constant data both sides are of the same order near the horizon.

Spatially constant supersolutions. Take constant envelopes $h_c \equiv \ell_c \geq \sup |g_c|$. If $\widehat{W}_c(t)$ does not depend on y and solves

$$\widehat{W}_c(t) = \vartheta \left[\ell_c + \sum_j |\kappa_j| \int_0^t \prod_i \widehat{W}_{c_j,i}(s) ds \right], \quad (4)$$

then it is a ϑ -supersolution, since $\int p_\sigma(y, z) dz = 1$. The branch step then factorizes: j and the children's remaining time $s = t - \sigma$ are drawn from $|\kappa_j| \prod_i \widehat{W}_{c_j,i}(s)$ on $(0, t)$, and $z = y + \sqrt{\sigma} \xi$ with a standard Gaussian ξ ; a leaf uses $z = y + \sqrt{t} \xi$ and returns $g_c(z)/\ell_c$. For polynomial f of degree m in Example 2.2, (4) is a finite ODE system in the codes $\text{Id}, F_0, \dots, F_m, D_1, \dots, D_d$. For $\vartheta = 1$ it is the sup-majorant ODE of [8, (3.10)] (there without gradient codes), which yields a sufficient integrability condition; here it defines the sampler.

Proposition 4.6 (Dimension-free sup majorant). *In Example 2.2 on $\mathbb{R}^d$ or $\mathbb{T}^d$ with polynomial f , let $\ell_{D_i} \geq \sup |\partial_i v|$ and $\ell_D = (\sum_i \ell_{D_i}^2)^{1/2}$. The minimal solution of (4) satisfies $\widehat{W}_{D_i} = (\ell_{D_i}/\ell_D) \widehat{W}_D$, where $(\widehat{W}_{\text{Id}}, \widehat{W}_{F_k}, \widehat{W}_D)$ solves the one-dimensional system with ℓ_D in place of $\sup |v'|$. Consequently the guaranteed horizon, the output bound and the work bound $\vartheta/(\vartheta - 1)$ depend on d only through ℓ_D , and the tilted sampler chooses the tuple (D_i, D_i, F_{k+2}) with direction i drawn with probability $\ell_{D_i}^2/\ell_D^2$.*

Proof. The equations for D_i are $\widehat{W}_{D_i} = \vartheta[\ell_{D_i} + \int_0^t \widehat{W}_{F_1} \widehat{W}_{D_i}]$, linear in ℓ_{D_i} given $\widehat{W}_{F_1}$, so the ansatz is consistent; and $\frac{1}{2} \sum_i \widehat{W}_{D_i}^2 = \frac{1}{2} \widehat{W}_D^2$ in the equations for F_k . Minimal solutions are unique for this finite locally Lipschitz system up to blow-up. The branch mass of the tuple (D_i, D_i, F_{k+2}) is the fraction $\ell_{D_i}^2/\ell_D^2$ of the aggregated one, and the D_i nodes have the same leaf and branch probabilities as the aggregated code, with leaf factor $\partial_i v(z)/\ell_{D_i}$. $\square$

For data depending on one direction, $v(x) = \psi(a \cdot x)$ with $|a| = 1$, one has $\ell_D = \sup |\psi'|$ for every d ; the same aggregation shows that the *exact* majorant W of the d -dimensional expansion equals the one-dimensional one, so the intrinsic horizon does not depend on d .

Corollary 4.7 (Certified confidence intervals). *Under Theorem 4.2(d), the mean $\bar{H}_n$ of n independent copies satisfies, by Hoeffding's inequality, $\mathbb{P}(|\bar{H}_n - U_{\text{Id}}(T, x)| \geq \widehat{W}_{\text{Id}}(T, x) \sqrt{2 \log(2/\alpha)/n}) \leq \alpha$.*

Remark 4.8 (Implementation). Our implementation tabulates the solution of (4) as a nondecreasing step function on an adaptive grid and computes every probability and density in the procedure from that same step function. Unbiasedness is then exact for the implemented sampler irrespective of the ODE error; only the slack is affected, and it is verified on the grid as $\vartheta_{\text{eff}} = \min_{c,i} \widehat{W}_c(s_i)/(\ell_c + \sum_j |\kappa_j| \int_0^{s_i} \prod \widehat{W})$ (the worst case in each cell is its right endpoint).

5 Explicit horizons

5.1 Constant data

Let $v \equiv r$ in Example 2.2, $a_k = |f^{(k)}(r)|$, $\Phi(z) = \sum_{k \geq 0} a_k z^k / k!$ with radius of convergence $R \in [0, \infty]$, and Z the solution of $Z' = \Phi(Z)$, $Z(0) = 0$.

Theorem 5.1 (Exact horizon). *Assume $a_0 > 0$ and put $\tau = \int_0^R dz / \Phi(z)$. Then $W_{F_k}(t) = \Phi^{(k)}(Z(t))$ and $W_{\text{Id}}(t) = |r| + Z(t)$ for $t < \tau$; $W_{\text{Id}}(\tau) = |r| + R$ if $\tau < \infty$; and $W_{\text{Id}}(t) = \infty$ for $t > \tau$. Moreover $W_{F_k}^\vartheta(t) = \vartheta \Phi^{(k)}(Z(\vartheta^2 t))$ and $W_{\text{Id}}^\vartheta(t) = \vartheta|r| + Z(\vartheta^2 t)$, so for every $T < \tau$ and $1 < \vartheta < \sqrt{\tau/T}$ Corollary 4.4 applies with $|H| \leq \vartheta|r| + Z(\vartheta^2 T)$ and $\mathbb{E}N_{\text{gen}} \leq \vartheta/(\vartheta - 1)$. If $a_0 = 0$ then $W_{\text{Id}} \equiv |r|$.*

Proof. Every tree containing a gradient code has a D -leaf (the D -chain $D \rightarrow (F_1, D)$ ends in a leaf with $g_D = 0$), so only the codes F_k and Id contribute, and (2) reduces to $V_k(t) = a_k + \int_0^t V_0 V_{k+1}$, $W_{\text{Id}}(t) = |r| + \int_0^t V_0$. Truncate the jet at order n (set $a_k = 0$ for $k > n$): this is a nonnegative truncation of tree weights, monotone in n and exhausting all finite trees. The truncated system has the minimal solution $V_k = \Phi_n^{(k)}(Z_n)$ with $Z'_n = \Phi_n(Z_n)$, $Z_n(0) = 0$, as one checks by differentiation; finite-dimensional uniqueness identifies it with the tree sum up to its explosion time $\tau_n = \int_0^\infty dz / \Phi_n$, after which the tree sum is infinite by monotonicity in t . As $n \rightarrow \infty$, $\Phi_n \uparrow \Phi$ on $[0, R)$ and $\Phi_n \uparrow \infty$ beyond R ; monotone convergence gives $\tau_n \downarrow \tau$ and $Z_n \uparrow Z$ on $[0, \tau)$. For $t > \tau$ some $\tau_n < t$, whence $W_{\text{Id}}(t) = \infty$. At $t = \tau$, grouping trees by the number of internal nodes shows $W_{\text{Id}}(t) - |r|$ is a power series in t with nonnegative coefficients, so its value at τ is the left limit R . For ϑ : $V_k = \vartheta \Psi_k(\vartheta^2 t)$ solves $V_k = \vartheta[a_k + \int_0^t V_0 V_{k+1}]$ iff Ψ solves the case $\vartheta = 1$, and minimality is preserved by the scaling; then $W_{\text{Id}}^\vartheta(t) = \vartheta|r| + \vartheta \int_0^t V_0 = \vartheta|r| + Z(\vartheta^2 t)$. If $a_0 = 0$ every tree through F_0 has a zero F_0 -leaf or a zero D -leaf. $\square$

Corollary 5.2 (Work near the horizon). *In Theorem 5.1, $W_N(T) = |r| + 2T \Phi(Z(T))$ for $T < \tau$. If Φ is a polynomial of degree $p \geq 2$, then as $T \uparrow \tau$:*

- (i) *every supported sampler with $|H| \leq C W_{\text{Id}}(T)$ a.s. has $\mathbb{E}N \geq (1 + o(1)) \frac{2\tau}{C(p-1)(\tau-T)}$;*
- (ii) *the tilt by W^ϑ with $\vartheta^2 = (\tau + T)/(2T)$ has $|H| \leq (2^{1/(p-1)} + o(1)) W_{\text{Id}}(T)$ and $\mathbb{E}N \leq (4\tau + o(1))/(\tau - T)$.*

Hence unbiased sampling with output bounded by a constant multiple of W_{Id} costs $\Theta((\tau - T)^{-1})$ expected nodes per sample.

Proof. $W_N = \partial_\vartheta[\vartheta|r| + Z(\vartheta^2 T)]_{\vartheta=1} = |r| + 2TZ'(T)$ by Theorem 3.1. Let Φ have leading term $a_p z^p / p!$. From $\int_{Z(t)}^\infty dz / \Phi = \tau - t$ one gets $Z(t)^{p-1} \sim p! / (a_p(p-1)(\tau - t))$ and $\Phi(Z)/Z \sim 1/((p-1)(\tau - t))$, so $W_N / W_{\text{Id}} \sim 2\tau / ((p-1)(\tau - T))$, and (i) follows from Theorem 3.1. For (ii), $\tau - \vartheta^2 T = (\tau - T)/2$, so $W_{\text{Id}}^\vartheta = \vartheta|r| + Z(\tau - (\tau - T)/2) \sim 2^{1/(p-1)} Z(T)$, and $\vartheta/(\vartheta - 1) \sim 2/(\vartheta^2 - 1) = 4T/(\tau - T)$. $\square$

Example 5.3. Allen–Cahn $f(u) = u - u^3$, $r = \frac{1}{2}$: $\Phi(z) = (z + \frac{3}{2})(z^2 + \frac{1}{4})$, $\tau = (3\pi - 2 \log 3)/5 \approx 1.44551$. For $f = \sin$, $r = \pi/2$: $\Phi = \cosh$, $\tau = \pi/2$ and $W_{\text{Id}}(\tau) = \infty$, while the ODE solution $2 \arctan e^t$ is global. For $f(y) = (1 + y^2)^{-1}$, $r = 0$: $\Phi(z) = (1 - z^2)^{-1}$, $R = 1$, $\tau = 2/3$ and $W_{\text{Id}}(2/3) = 1 < \infty$, but $W_{\text{Id}}(t) = \infty$ for every $t > 2/3$.

5.2 Fixed-rate samplers

Proposition 5.4. *In the setting of Theorem 5.1, consider the fixed-rate sampler with exponential lifetimes of rate $\lambda > 0$ and probability $p \in (0, 1]$ on the tuple (F_0, F_{k+1}) (the other tuple carries zero weight). Let $\Phi_2(z) = \sum_k a_k^2 z^k / k!$ and $\tau_2 = \int_0^{R_2} dz / \Phi_2$. If Φ_2 is a polynomial of degree at least two, then $\mathbb{E}H^2 < \infty$ iff*

$$T < T_2(\lambda, p) = \lambda^{-1} \log(1 + \lambda^2 p \tau_2), \quad \sup_{\lambda > 0} T_2(\lambda, p) = K \sqrt{p \tau_2}, \quad K = \max_{z > 0} \frac{\log(1+z)}{\sqrt{z}} \approx 0.804743.$$

For Example 5.3, $\tau_2 = 1.530121 \dots$ and $\sup_{\lambda, p} T_2 = 0.99545 < \tau$.

Proof. A leaf at remaining time t has weight $g/e^{-\lambda t}$, a branch $\kappa/(p\lambda e^{-\lambda \sigma})$. Conditioning on the first event, the second moments M_c satisfy $M_{F_k}(t) = a_k^2 e^{\lambda t} + (p\lambda)^{-1} \int_0^t e^{\lambda \sigma} M_{F_0}(t - \sigma) M_{F_{k+1}}(t - \sigma) d\sigma$ (minimal solutions, as in Lemma 2.4). Since the tuples are binary, writing $M_{F_k} = e^{\lambda t z_k}$ and changing time to $\theta = (e^{\lambda t} - 1)/\lambda^2$ (so $d\theta = \lambda^{-1} e^{\lambda t} dt$) gives

$$z_k(\theta) = a_k^2 + p^{-1} \int_0^\theta z_0 z_{k+1} d\theta'.$$

Introduce s by $ds/d\theta = z_0$, $s(0) = 0$. Then $dz_k/ds = z_{k+1}/p$, so $z_k = \Phi_2^{(k)}(s/p)$ (a finite sum), and $v = s/p$ solves $dv/d\theta = \Phi_2(v)/p$. Hence the F -moments are finite exactly for $\theta < \theta_* = p\tau_2$ and diverge as $\theta \uparrow \theta_*$; as in Theorem 5.1 the finite polynomial system identifies the minimal solution, and monotonicity in t gives divergence after θ_* . The single-child Id code has $M_{\text{Id}}(t) = e^{\lambda t} [r^2 + \lambda^{-1} \int_0^t z_0 dt']$ and $\int z_0 dt' = \int \lambda e^{-\lambda t'} ds \geq \lambda(1 + \lambda^2 \theta_*)^{-1} s \rightarrow \infty$ as $\theta \uparrow \theta_*$, so the root diverges at the same time. Solving $\theta_* = (e^{\lambda T} - 1)/\lambda^2$ gives T_2 , and with $z = \lambda^2 p \tau_2$, $T_2 = \sqrt{p\tau_2} \log(1+z)/\sqrt{z}$, maximized at the unique root of $2z/(1+z) = \log(1+z)$, $z_* \approx 3.9215$. $\square$

Corollary 5.5 (Uniform tuples in dimension d). *In $\mathbb{R}^d$, the mechanism of Example 2.2 gives each F_k the $1 + d$ tuples (F_0, F_{k+1}) and (D_i, D_i, F_{k+2}) , $i \leq d$. For constant data the latter carry zero weight, so a fixed-rate sampler with uniform tuple choice, as in [18, 19], is the case $p = 1/(d+1)$ of Proposition 5.4: its variance is finite exactly for $T < \lambda^{-1} \log(1 + \lambda^2 \tau_2/(d+1))$, and $\sup_\lambda T_2 = K \sqrt{\tau_2/(d+1)}$. For flat Allen–Cahn data in $d = 100$ this is 0.0990 (0.0294 at $\lambda = 2$), against the dimension-independent intrinsic horizon 1.4455.*

Remark 5.6. For a two-term series $\Phi(z) = a_0 + a_p z^p / p!$ the substitution $z = (a_0 p! / a_p)^{1/p} u$ gives $\tau = a_0^{-1} (a_0 p! / a_p)^{1/p} I_p$ and $\tau_2 = a_0^{-2} (a_0^2 p! / a_p^2)^{1/p} I_p$, $I_p = \int_0^\infty du / (1 + u^p) = \frac{\pi/p}{\sin(\pi/p)}$. Hence the best fixed-rate sampler reaches the fraction

$$\frac{\sup_{\lambda, p} T_2}{\tau} = K (p!)^{-1/(2p)} I_p^{-1/2},$$

independently of a_0, a_p : 0.540 for $p = 2$, 0.543 for $p = 3$, 0.454 for $p = 6$, 0.375 for $p = 10$, and $\sim K \sqrt{e/p} \rightarrow 0$ as $p \rightarrow \infty$ (checked against quadrature to 10^{-15}). Fixed-rate samplers thus lose a fraction of the intrinsic horizon that grows with the degree of the nonlinearity, whereas Theorem 4.2 attains all of it.

5.3 Compact tori

Theorem 5.7. *Let $\mathbf{X} = \mathbb{T}^d$, v smooth and nonconstant, f real-analytic on an open interval containing $v(\mathbf{X})$, in Example 2.2. Then $W_{\text{Id}}(T, x) < \infty$ for every T and x iff f is affine on $v(\mathbf{X})$. More*

generally, for smooth f and smooth v , all-horizon integrability holds iff $f(v) \equiv 0$ or $f^{(p)}(v) \equiv 0$ for all $p \geq 2$; if $f(v) \not\equiv 0$ and $f^{(p)}(v) \not\equiv 0$, then with $\eta_q = \min_x P_{t_0}|f^{(q)}(v)|(x) > 0$,

$$W_{\text{Id}}(T, x) = \infty \quad \text{for all } x \text{ and } T \geq t_0 + \frac{p}{p-1} \eta_0^{-(p-1)/p} (p!/\eta_p)^{1/p}.$$

Proof of the obstruction. Positivity of the heat kernel on the compact torus makes η_0, η_p positive. Split the time integral in (2) at t_0 , drop the (nonnegative) part before t_0 and all tuples except the coefficient-one chain $F_k \rightarrow (F_0, F_{k+1})$. The functions $A_k(s) = \inf_x W_{F_k}(t_0 + s, x)$ satisfy $A_k(s) \geq \eta_k \mathbf{1}_{k \in \{0, p\}} + \int_0^s A_0 A_{k+1}$ for $k < p$ and $A_p \geq \eta_p$, since $P_{t_0+s}|g| = P_s(P_{t_0}|g|) \geq \min P_{t_0}|g|$. Comparison with the finite system $V'_k = V_0 V_{k+1}$ ($k < p$), $V_p = \eta_p$ gives $A_0 \geq V_0 = \eta_0 + (\eta_p/p!)Z^p$ with $Z' = V_0$, $Z(0) = 0$, which explodes before $\int_0^\infty dz/(\eta_0 + \eta_p z^p/p!)$; splitting this integral at $(\eta_0 p!/\eta_p)^{1/p}$ gives the displayed bound. The comparison is carried out on finite Picard iterates, so no infinite moment is differentiated; $W_{\text{Id}} \geq \int_0^T P_{T-s} W_{F_0}(s) ds$ transfers divergence to the root. $\square$

Proof of sufficiency. See Appendix B. $\square$

5.4 A Fujita-type threshold on $\mathbb{R}^d$

Theorem 5.8. Let $\mathbf{X} = \mathbb{R}^d$, $f(u) = -u^2$ and $v = \varepsilon e^{-|x|^2}$ with $0 < \varepsilon \leq 1/32$. Then $W_{\text{Id}}(T, x) < \infty$ for all T, x iff $d \geq 5$. For $d \geq 5$, $W_{\text{Id}}(t, x) \leq (\varepsilon + C\varepsilon^2 t)G(t, x)$ with $G(t, x) = (1+t)^{-d/2} e^{-|x|^2/(2(1+t))}$, $C = 2 + 32d/(d-2)$, and Theorem 4.2 applies with the supersolution $(aG, bG, \delta G, (\varepsilon + C\varepsilon^2 t)G)$ defined below, for every horizon.

Proof for $d \geq 5$. Write A, B, D_i for the majorants of F_0, F_1, D_i ($W_{F_2} = 2$, higher codes vanish). Then (2) is the mild form of $(\partial_t - L)A = AB + \sum_i D_i^2$, $(\partial_t - L)B = 2A$, $(\partial_t - L)D_i = BD_i$, $(\partial_t - L)W_{\text{Id}} = A$ with data $v^2, 2v, |\partial_i v|, v$. With $w(t) = (1+t)^{-d/2}$ one has $(\partial_t - L)G = 0$, $G^2 \leq wG$, $I_0 = \int_0^\infty w = 2/(d-2) \leq 2/3$, $I_1 = \int_0^\infty tw = 4/((d-2)(d-4)) \leq 4/3$. Put $b(t) = 2\varepsilon + 2C\varepsilon^2 t$, $J(t) = \int_0^t bw$, $\delta(t) = 2\varepsilon e^{J(t)}$ and $a(t) = \varepsilon^2 + C\varepsilon^2 J(t) + 4d\varepsilon^2 e^{2J(\infty)} \int_0^t w$. Then $J(\infty) \leq 107/576 < 1/4$, $a \leq \frac{3}{4}C\varepsilon^2$, and $a' \geq w(ab + d\delta^2)$, $b' \geq 2a$, $\delta' = w b \delta$; since each product contributes $G^2 \leq wG$, $(aG, bG, \delta G)$ is a supersolution dominating the data ($v \leq \varepsilon G(0)$, $v^2 \leq \varepsilon^2 G(0)$, $|\partial_i v| \leq 2\varepsilon G(0)$). $\square$

Proof for $d \leq 4$. See Appendix C. The reduced system is a mixed-exponent system in the scope of [7]; the appendix gives a self-contained argument with explicit horizons. $\square$

6 Numerical experiments

All experiments use Example 2.2 in $d = 1$ with $f(u) = u - u^3$, the spatially constant supersolution (4), and pure-Python samplers; the code is available at [TODO: repository].

Flat data. For $v \equiv \frac{1}{2}$ the tabulated bound agrees with $\vartheta/2 + Z(\vartheta^2 T)$ to the printed digits at $T = 0.5, 0.9, 1.2$. At $T = 1.4$ (96.9% of τ , beyond every fixed-rate sampler by Proposition 5.4), $\vartheta = 1.004$ and 1.6×10^6 samples give 0.917704 ± 0.002741 against the exact value 0.919628, with $|H| \leq 3.847$ and about 33 nodes per sample. The sampler-independent identity $\mathbb{E}[N|H|] = W_N = \frac{1}{2} + 2T\Phi(Z(T))$ of Theorem 3.1 is reproduced within one standard error (2×10^5 samples): 2.226 ± 0.007 vs 2.219 at $T = 0.9$, 7.487 ± 0.030 vs 7.512 at $T = 1.2$, 101.5 ± 0.4 vs 101.85 at $T = 1.4$; the measured $\mathbb{E}N$ (2.51, 6.21, 32.9) lies between the lower bound W_N/M (1.75, 3.61, 26.5) and $\vartheta/(\vartheta - 1)$ (11, 21, 251).

T	reference	tilted (s.e.)	Hoeffding 95%	rate 1 (s.e.), max $ H $	rate 2 (s.e.), max $ H $
0.5	0.68946	0.68884 (0.00051)	0.0046	0.68905 (0.00075), 23	0.68932 (0.00127), 10
0.8	0.78916	0.78716 (0.00112)	0.0061	0.78626 (0.00411), 665	0.79102 (0.00267), 108
1.0	0.84334	0.84247 (0.00186)	0.0081	0.82450 (0.01567), 2630	0.85240 (0.00873), 854
1.1	0.86632	0.87069 (0.00253)	0.0104	0.91542 (0.02545), 4650	0.79822 (0.05822), 11200
1.2	0.88660	0.88633 (0.00407)	0.0178	0.94836 (0.01935), 1420	0.93548 (0.06933), 9330

Table 1: Periodic Allen–Cahn, $v = \frac{1}{2} + 0.02 \sin(2\pi x)$, $x = \frac{1}{4}$. Tilted sampler with $\vartheta = 1.02$ (1.0169 at $T = 1.2$); mean nodes per sample 1.5, 2.2, 3.9, 6.4, 16.4. Fixed-rate standard errors at $T \geq 1$ are not reliable (possibly infinite variance).

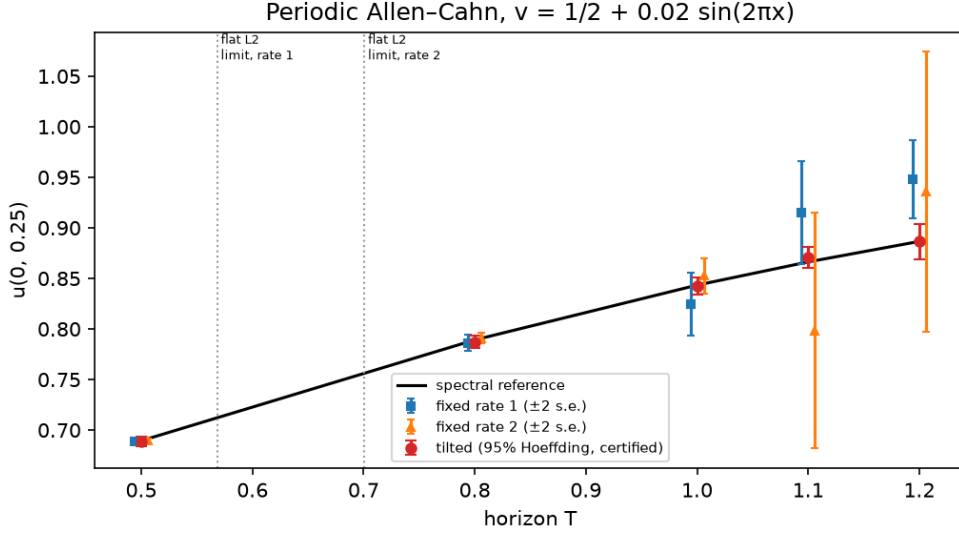


Figure 1: Estimates of $u(0, \frac{1}{4})$ against the horizon. Vertical lines: finite-variance limits of the fixed-rate samplers for the flat data (Proposition 5.4).

Periodic data. For $v = \frac{1}{2} + 0.02 \sin(2\pi x)$ on $\mathbb{T}^1$ the exact sup envelopes give the guaranteed horizon $\tau_{\text{sup}} = 1.28237$. Figure 1 and Table 1 compare the tilted sampler with fixed-rate samplers (rates 1 and 2, uniform tuples) at $x = \frac{1}{4}$, 2×10^5 samples each, against a pseudo-spectral reference (256 modes, Strang splitting with exact reaction flow; step halving changes it by less than 1.4×10^{-11}). The tilted estimates lie within 1.8 standard errors of the reference and inside the certified Hoeffding intervals at every T . The fixed-rate samplers exhibit heavy tails from $T \approx 0.8$ on: the largest 0.1% of samples carry up to 18% of $\sum |H|$, maximal $|H|$ reach 10^4 , and at $T = 1.2$ the rate-one estimate deviates by 3.2 of its own standard errors.

Dimension 100. For $v(x) = \frac{1}{2} + 0.02 \sin(2\pi a \cdot x)$ on $\mathbb{R}^{100}$ with $a = (1, \dots, 1)/10$, the exact solution is the one-dimensional periodic solution evaluated at $a \cdot x$, since $a \cdot B$ is a standard Brownian motion; all samplers nevertheless simulate the 100-dimensional tree. By Proposition 4.6 the tilted sampler has the same guaranteed horizon 1.28237 as in one dimension. We compare it at $x_0 = a/4$ with fixed-rate samplers ($\lambda = 2$) using uniform tuples over the 101 tuples of each F_k code, and a tuned main-label probability 0.95 (Table 2, Figure 2, 10^5 samples each). The flat-data limits of Corollary 5.5 are 0.029 (uniform) and 0.96 (tuned). The uniform sampler is heavy-tailed from

T	reference	tilted (s.e.), nodes	uniform $\lambda = 2$ (s.e.), $\max H $	tuned $\lambda = 2$ (s.e.), $\max H $
0.1	0.54072	0.54128 (0.00025), 1.07	0.54114 (0.00052), 4.0	0.54056 (0.00050), 0.64
0.3	0.61476	0.61510 (0.00040), 1.23	0.61643 (0.00120), 8.9	0.61465 (0.00108), 0.95
0.6	0.72479	0.72566 (0.00094), 1.64	0.68449 (0.05882), 5876	0.72528 (0.00214), 3.7
1.0	0.84334	0.83839 (0.00265), 3.90	0.93657 (0.01059), 853	0.84578 (0.00503), 62
1.2	0.88660	0.86767 (0.00576), 16.6	1.03702 (0.00934), 190	0.88252 (0.02646), 2030

Table 2: Allen–Cahn in $d = 100$, $x_0 = a/4$, one seed per column; Hoeffding 95% half-widths of the tilted sampler: 0.0049, 0.0057, 0.0071, 0.0115, 0.0252.

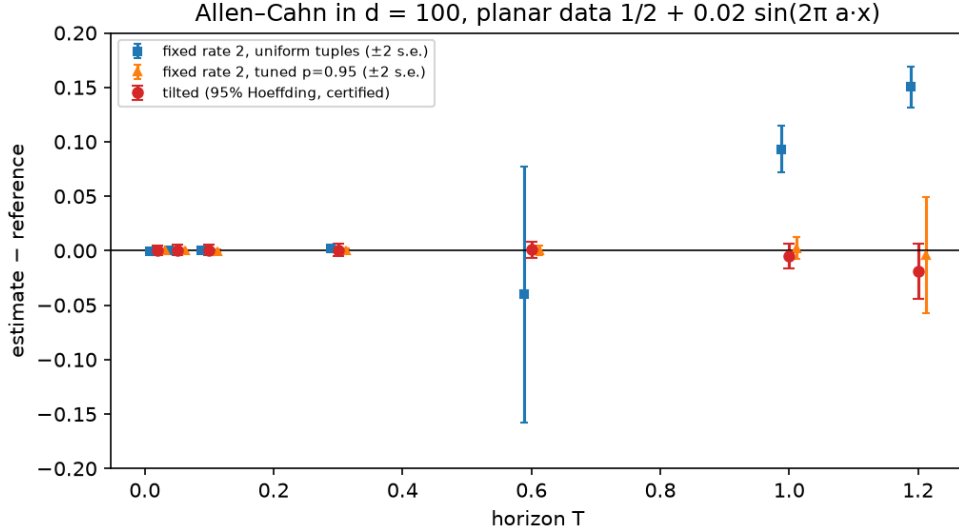


Figure 2: Error of the estimates of $u(0, a/4)$ in dimension 100.

$T = 0.6$ ($\max |H| = 5876$) and wrong by 8.8 and 16 of its standard errors at $T = 1.0$ and 1.2; the tuned sampler holds to $T = 1.0$ and turns heavy-tailed at $T = 1.2$ ($\max |H| = 2030$); the tilted sampler’s certified intervals cover the reference at every T . Independent repetitions of the tilted runs (ten seeds, up to 2×10^6 samples) give pooled z -scores -0.03 , -0.25 , -0.39 , $+1.56$ at $T = 0.02, 0.1, 1.0, 1.2$.

A limitation. For the traveling-wave data $v(x) = -\frac{1}{2} - \frac{1}{2} \tanh(-x/2)$ of [18], the range of v is $(-1, 0)$ and the sup envelopes combine the jets of f near -1 and near 0 ; the guaranteed horizon is only 0.666, and within it the rate-one fixed sampler has a two to three times smaller product of variance and work (with unbounded outputs). Spatially varying supersolutions, as in Theorem 5.8, are needed for such data.

7 Discussion

The results separate three quantities that are often conflated: the existence horizon of the PDE, the integrability horizon of a fixed expansion, and the finite-variance horizon of a particular sampler. The second is characterized by one positive system and cannot be improved by sampling; the gap between the second and the third can be closed completely, with bounded output and horizon-free

expected work, whenever a supersolution with slack is available. Extending the horizon beyond the integrability horizon requires changing the expansion itself (e.g. time-patching [12, 21] or bounded voting representations), which is outside the scope of this paper. Open problems include computable spatially varying supersolutions for general data, the fully nonlinear mechanisms of [18], and the endpoint work frontier.

A Proof of Proposition 2.5

Let $m = \deg f$. Codes F_k with $k > m$ have $g_{F_k} \equiv 0$ and every tuple issued from them contains a code $F_{k'}$ with $k' > k$; by induction on the depth, every finite tree rooted at such a code has $a = 0$. The remaining codes $\text{Id}, F_0, \dots, F_m, D_1, \dots, D_d$ form a finite set, and by Lemma 2.4 the vector $Y = (U_c)_c$ is a mild solution on $[0, T]$ of the finite signed system

$$Y_c(t) = P_t g_c + \sum_j \kappa_j \int_0^t P_{t-s} \left[\prod_i Y_{c_{j,i}}(s) \right] ds, \quad (5)$$

with $|Y_c| \leq W_c \leq M$ on $[0, T] \times \mathbf{X}$.

By standard semilinear parabolic theory in $BUC(\mathbf{X})$ (locally Lipschitz f), (1) has a unique mild solution u on a maximal interval $[0, T_{\max})$, with $\limsup_{t \uparrow T_{\max}} \|u(t)\|_\infty = \infty$ if $T_{\max} < \infty$. For $v \in C_b^2$ and polynomial f , u is classical on $(0, T_{\max})$, and on every $[0, t']$ with $t' < T_{\max}$ the functions $u, \nabla u$ are bounded and continuous up to $t = 0$; for ∇u this follows from $\nabla u(t) = P_t \nabla v + \int_0^t \nabla P_{t-s} f(u(s)) ds$ and $\|\nabla P_r\|_{\infty \rightarrow \infty} \leq Cr^{-1/2}$. The chain rule gives

$$(\partial_t - L)f^{(k)}(u) = f^{(k+1)}(u)f(u) - \frac{1}{2}f^{(k+2)}(u)|\nabla u|^2, \quad (\partial_t - L)\partial_i u = f'(u)\partial_i u,$$

so by Duhamel's formula for bounded classical solutions the physical family $\tilde{Y} = (u, f^{(k)}(u), \partial_i u)$ solves (5) on $[0, t']$ with the same initial data.

The right-hand side of (5) is a polynomial map, Lipschitz on bounded subsets of $BUC(\mathbf{X})^{|C|}$ with a constant Λ . For $t' < \min(T, T_{\max})$ both Y and $\tilde{Y}$ are bounded on $[0, t']$, and $\|Y(t) - \tilde{Y}(t)\|_\infty \leq \Lambda \int_0^t \|Y(s) - \tilde{Y}(s)\|_\infty ds$ with P_{t-s} a contraction; Gronwall's lemma gives $Y = \tilde{Y}$. In particular $\|u(t)\|_\infty = \|U_{\text{Id}}(t)\|_\infty \leq M$ on $[0, \min(T, T_{\max}))$, so the blow-up alternative excludes $T_{\max} \leq T$. Hence $U_{\text{Id}} = u$ on $[0, T]$. $\square$

B Proof of Theorem 5.7: sufficiency and the classification

Throughout, “tree” means a finite completed tree of Example 2.2, and $\mathbf{X}$ is connected.

(i) *Higher jets vanish.* Suppose $f^{(p)}(v) \equiv 0$ for all $p \geq 2$. We claim $a(\theta) = 0$ for every tree rooted at F_k , $k \geq 2$. Induct on the depth: a leaf has $g_{F_k} = f^{(k)}(v) \equiv 0$; a branch uses (F_0, F_{k+1}) or (D_i, D_i, F_{k+2}) , each containing a child $F_{k'}$ with $k' \geq 3$, whose subtree has zero weight. Hence $W_{F_k} \equiv 0$ for $k \geq 2$, and every tuple containing such a code contributes nothing. Moreover $\nabla f'(v) = f''(v)\nabla v \equiv 0$, so $f'(v) \equiv b$ is constant on the connected space $\mathbf{X}$. The surviving part of (2) reads $W_{F_1} = P_t |b| = |b|$ and $W_{F_0}(t) = P_t |f(v)| + |b| \int_0^t P_{t-s} W_{F_0}(s) ds$, whose minimal solution is $e^{|b|t} P_t |f(v)|$. Therefore

$$W_{\text{Id}}(t) = P_t |v| + \frac{e^{|b|t} - 1}{|b|} P_t |f(v)| \quad (\text{with } t \text{ in place of the fraction if } b = 0),$$

which is finite for all t .

(ii) $f(v) \equiv 0$. If v is nonconstant, $v(\mathbf{X})$ is a nondegenerate interval on which f vanishes, hence all derivatives $f^{(k)}$ vanish on it and $g_{F_k} \equiv 0$ for every k . Every tree rooted at some F_k has zero weight by induction on the depth (a leaf is zero; a branch contains an F -child). Thus $W_{F_0} \equiv 0$ and $W_{\text{Id}} = P_t|v|$. If $v \equiv r$ with $f(r) = 0$, then $g_{F_0} = 0$ and $g_{D_i} = 0$; every tree rooted at D_i has zero weight (each branch $D_i \rightarrow (F_1, D_i)$ contains a D_i -child), and every tree rooted at F_0 has zero weight (a branch contains an F_0 -child or a D_i -child). Again $W_{\text{Id}} = |r|$.

(iii) *Necessity*. If neither (i) nor (ii) holds, $f(v) \not\equiv 0$ and $f^{(p)}(v) \not\equiv 0$ for some $p \geq 2$, and the obstruction proved in Section 5 applies. For completeness we record the comparison step. The finite system $V_k = \eta_k \mathbf{1}_{k \in \{0\}} + \int_0^s V_0 V_{k+1}$ ($0 \leq k < p$), $V_p \equiv \eta_p$, has $V_{p-j} = \eta_p Z^j / j!$ for $1 \leq j \leq p-1$ and $V_0 = \eta_0 + \eta_p Z^p / p!$ with $Z' = V_0$, $Z(0) = 0$; its Picard iterates started at 0 are dominated by (A_k) by induction, since A satisfies the same inequalities. Hence $A_0 = \infty$ after the explosion time $s_* = \int_0^\infty dz / (\eta_0 + \eta_p z^p / p!)$, and splitting the integral at $z_1 = (\eta_0 p! / \eta_p)^{1/p}$ gives $s_* < z_1 / \eta_0 + p! / ((p-1)\eta_p z_1^{p-1}) = \frac{p}{p-1} z_1 / \eta_0$, the bound displayed in Theorem 5.7. Since $W_{\text{Id}}(T, x) \geq \int_0^T P_{T-r} W_{F_0}(r)(x) dr$ and $W_{F_0} = \infty$ on $(t_0 + s_*, \infty) \times \mathbf{X}$, $W_{\text{Id}}(T, x) = \infty$ for $T > t_0 + s_*$.

(iv) *Classification*. (i)–(iii) prove the smooth statement. If f is real-analytic on an open interval $J \supset v(\mathbf{X})$ and v is nonconstant, then $f'' \equiv 0$ on the nondegenerate interval $v(\mathbf{X})$ implies f affine on J by the identity theorem, and $f(v) \equiv 0$ implies $f \equiv 0$ on J ; conversely an affine f satisfies (i). For constant $v \equiv r$ the condition reads: $f(r) = 0$, or f is affine on J . $\square$

C Proof of Theorem 5.8 for $d \leq 4$

Discard the nonnegative source $\sum_i D_i^2$ and let A_*, B_* be the minimal extended solutions of

$$A_* = P_t A_0 + \int_0^t P_{t-s}(A_* B_*) ds, \quad B_* = P_t B_0 + 2 \int_0^t P_{t-s} A_* ds,$$

$A_0 = \varepsilon^2 e^{-2|x|^2}$, $B_0 = 2\varepsilon e^{-|x|^2}$; then $W_{F_0} \geq A_*$ and $W_{F_1} \geq B_*$. All Picard iterates, hence A_*, B_* , are radial and radially nonincreasing (the heat semigroup, products and sums preserve this).

Correlation. For nonnegative radial nonincreasing ϕ, ψ and $\ell > 0$, $P_\ell(\phi\psi)(0) \geq P_\ell\phi(0)P_\ell\psi(0)$ (Chebyshev's inequality for two nonincreasing functions of the radius of a centred Gaussian; truncation and monotone convergence cover extended values).

Riccati barrier. Fix $t_0 \geq 0$, $T > 0$, and set $a(s) = P_{T-s} A_*(t_0 + s)(0)$, $b(s) = P_{T-s} B_*(t_0 + s)(0)$ for $0 \leq s < T$. Splitting the Volterra integrals at t_0 and using the correlation inequality, $a(s) \geq a(0) + \int_0^s ab$ and $b(s) \geq 2 \int_0^s a$. With $z(s) = \int_0^s a$ this gives $z' \geq \alpha + z^2$ for $\alpha = a(0)$, so z is infinite before $\pi/(2\sqrt{\alpha})$. If $\pi/(2\sqrt{\alpha}) < T$ the Id integral at horizon $t_0 + T$ diverges at the origin. It diverges at every x because $P_\ell h(x) \geq e^{-|x|^2/(2\ell)} P_\ell h(0)$ for even nonnegative h (pair y with $-y$), with $\ell \geq T - \pi/(2\sqrt{\alpha}) > 0$.

$d \leq 3$. With $t_0 = 0$, $a(0) = \varepsilon^2(1 + 4T)^{-d/2}$, and the condition $T > \frac{\pi}{2\varepsilon}(1 + 4T)^{d/4}$ holds for $T \geq T_d = \max\{1, (\pi 5^{d/4}/\varepsilon)^{4/(4-d)}\}$, since $1 + 4T \leq 5T$ for $T \geq 1$. Hence $W_{\text{Id}}(t, x) = \infty$ for $t \geq T_d$.

$d = 4$. From the equations, $A_*(t) \geq P_t A_0$ and $B_*(t) \geq 2t P_t A_0$, so $A_*(t) \geq L_A(t) := P_t A_0 + \int_0^t P_{t-s}[2s(P_s A_0)^2] ds$, a Gaussian mixture of total mass

$$M_L(t) = \frac{\varepsilon^2 \pi^2}{4} + \frac{\varepsilon^4 \pi^2}{128} \left[\log(1 + 4t) + \frac{1}{1 + 4t} - 1 \right],$$

every component of which has variance at most $t + \frac{1}{4}$. Let $K = 1 + 512\pi^2/\varepsilon^4$, $t_0 = (e^K - 1)/4$ and $T = t_0 + \frac{1}{4}$. Then $M_L(t_0) > 4\pi^4$ and, since a unit-mass Gaussian of variance $q \leq T$ has P_T -value at the origin at least $(4\pi T)^{-2}$, $a(0) \geq P_T L_A(t_0)(0) \geq M_L(t_0)/(16\pi^2 T^2) > \pi^2/(4T^2)$. The Riccati barrier gives $W_{\text{Id}}(t, x) = \infty$ for all $t \geq t_0 + T = (2e^K - 1)/4$ and all x . $\square$

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